Executive Research Portfolio & Extended Abstract: Machine-Checked Robotics, Control Theory, Experimental Mathematics, and Combinatorics in Lean 4

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Abstract

This executive portfolio synthesizes four publication tracks developed across our local and DGX Spark (`sparky.local`) formalization and computational mathematics infrastructure. Every theorem reported in the Lean 4 tracks is kernel-checked with **zero `sorry`**, **zero `native_decide`**, and only the three standard Lean 4 axioms (`propext`, `Classical.choice`, `Quot.sound`), while every computational certificate is verified in exact rational arithmetic ($\mathbb{Q}$) with zero floating-point residual.

1. Comparative Overview of the Four Manuscript Tracks

2. Key Visual Highlights Across Tracks

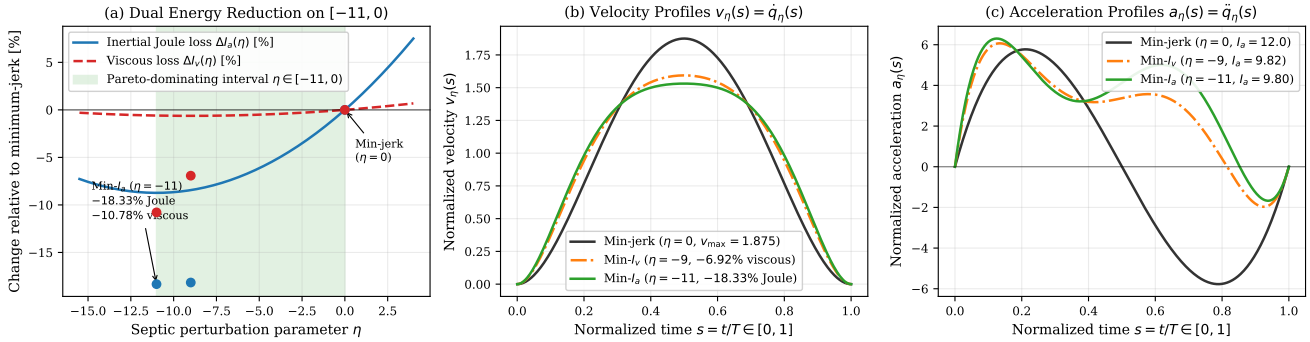


Figure 1: **Track 1 (Robotics & Control)**: Exact polynomial antiderivative verification in `RobotEnergySmoothness.lean` showing that every septic parameter $\eta \in [-11, 0)$ strictly reduces both inertial I^2R Joule loss $\int_0^T \dddot{q}^2\, dt$ (up to -18.33% at $\eta = -11$) and viscous dissipation $\int_0^T \ddot{q}^2\, dt$ (up to -10.79% at $\eta = -11$) relative to classical minimum-jerk ($\eta = 0$).

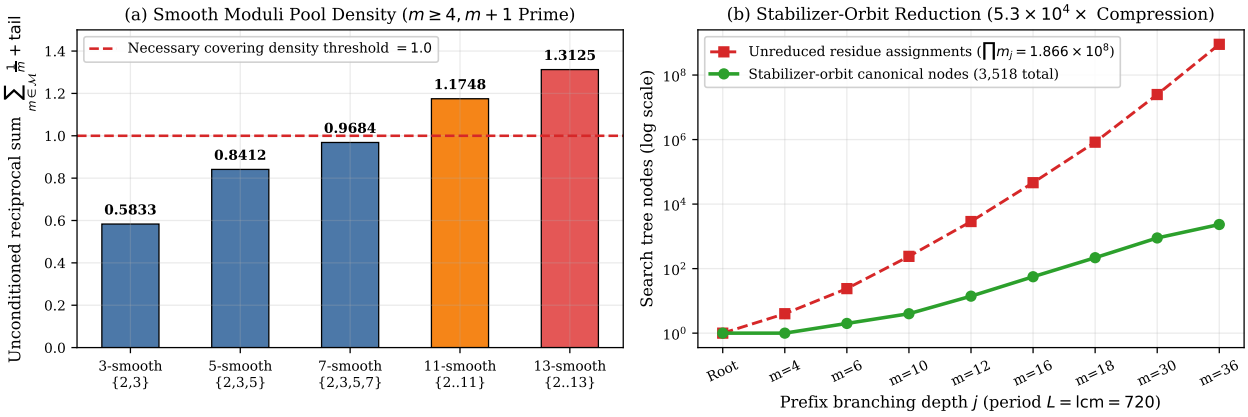


Figure 2: **Track 3 (Experimental Math — Erdős 273)**: Unconditioned reciprocal density across k-smooth prime pools ($m \geq 4, m+1$ prime) and the $5.3 \times 10^4\times$ compression achieved by stabilizer-orbit canonical branching modulo $L = \mathrm{lcm}(P) = 720$.

Track / Draft	Primary Mathematical & Engineering Contributions	Core Verified Artifacts	Evidence Tier	Target Venue
Paper 1: Robotics & Control in Lean 4	1. Septic trajectory family strictly Pareto-dominates minimum-jerk ($\eta = 0$) on $\eta \in [-11, 0)$ for both I^2R Joule loss (-18.33%) and viscous loss (-10.79%). 2. Clock-slew rate step $\alpha_- \neq \alpha_+$ destroys C^2 spline differentiability at knots when $v_k \neq 0$. 3. 3D 4-contact bilinear ZMP normal load + Lorentz friction cone + ODE forward invariance. 4. Coupled 6-DOF $SE(3)$ homography visual servoing depth Lie-derivative rotational cancellation	`RobotEnergySmoothness.lean` `WorldFrameTransport.lean` `LorentzConeCBF.lean` `VisualServoingInvariant.lean` `HomographyDecomposition.lean` + 26 `ControlsLean` & 19 `RoboLean` modules	**Tier 1**: 100% Kernel-Checked Lean 4 (0 `sorry`)	*IEEE RA-L / L-CSS* or *ACM CPP / ITP*
Paper 2: Byers–Nash ($n = 6$) Duality & Normal Form	1. Unconditional Lean 4 proof of 6×6 rank-≤ 4 skew normal form and same-square Lyapunov exclusion. 2. Common-T involution proof that $s(A) \geq 1$ across the 8D equality locus. 3. Exact 9D Sylvester reduction & 21×21 rational $LDL^\top \succ 0$ dual SDP certificates.	`SkewRankFourNormalForm.lean` `ByersNashDuality.lean` `certify_mf13_pair.py`	**Tier 1 + Tier 2**: Lean 4 Theorems + Exact $\mathbb{Q}$ SDP	*SIMAX / Linear Algebra Appl. (LAA)*
Paper 3: Experimental Math (Erdős #273 & #885)	1. Excludes 11-smooth covering systems with $m + 1$ prime (5,265 moduli, unconditioned sum $1.1748 > 1$) via 3,518-node stabilizer-orbit tree ($5.3 \times 10^4\times$ compression, min survivor $\lvert U\rvert = 212/720$, gap $> 4.08 \times 10^{-5}$) + Lean 4 Euler tail bounds. 2. Proves 5 quadrics in $\mathbb{P}^6$ for 6×4 Erdős 885 grids form a smooth complete intersection of genus $g = 49$, and proves via Mazur's torsion theorem that all 20 elliptic quotients E_{ijk} have $\mathrm{rank}(E_{ijk}(\mathbb{Q})) \geq 1$ (blocking plain Chabauty).	`CoveringEleven.lean` `work/273/check_eleven.py` `verify_885_geometry.py` `Erdos273.lean`, `Erdos885.lean`	**Tier 1 + Tier 2**: Lean 4 Tail & Variant Proofs + Audited $\mathbb{Q}$ Certificates	*Experimental Mathematics / Math. Comp.*
Paper 4: Formalized Combinatorics & Algebra	1. Erdős Problem 348: complete Lean 4 proof of Geneson's deletion-threshold classification ($m \leq 1$). 2. Erdős Problem 216 ($k \geq 7$): full Euclidean formalization of Horton sets avoiding empty convex 7-gons. 3. Erdős Problem 609 ($n = 3$): kernel-checked proof that $f(3) = 5$ on K_9. 4. Kourovka 17.32 & 19.35, MO 513215 / 509929 / 515611 / 501066, plus 6 Erdős variants in `ErdosLab`.	`erdos-348-lean` (1,180 LOC) `lean-erdos-proofs` (2,380 LOC) `ctrl/lean/ErdosLab` (6 problems)	**Tier 1**: 100% Kernel-Checked & Comparator-Audited	*ITP / CPP / Journal of Automated Reasoning*